

Storing and retrieving data

All programs, from simple calculators to flashy websites, have to store and manipulate data. The most basic programming tools used for achieving this are variables, constants, and arrays.

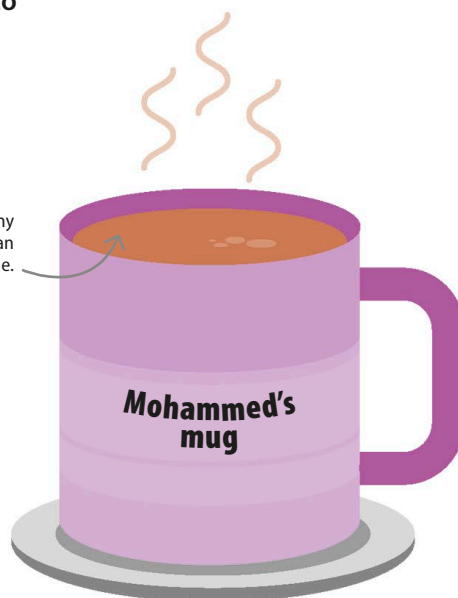
Variables

Variables are a storage mechanism, similar to a mug with a name, value, and size. Just as the contents of a mug can change over time, the values inside a variable can change during the course of a program. Also like a mug, each variable has a specific size. It's important to pick the right variable size from the start as different kinds of data stored in the variable can take up varying amounts of space in the memory.

► Storing variables

Variables are always labelled. To search and access data easily, programmers choose helpful, descriptive names for their variables.

The mug can hold many different things, but it can only hold one at a time.



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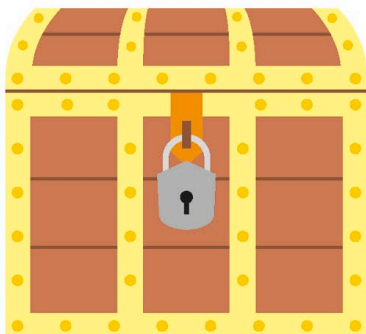
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Constants

Constants work like variables, but they're used for data that can't change while a program is running (if ever). They are useful when modifying programs, and are used to store tricky mathematical values (such as the value of Pi), scientific constants, or application-specific values. Anything that could be forgotten, misspelled, or changed only once every few years is best stored in a constant.

► Fixed value

Once a value is assigned to a constant, its value can't be changed unless the program is stopped and restarted. This is similar to a locked chest, whose contents can only be removed if it is unlocked.



REAL WORLD

Company banner

Let's say a library is rebranding its website. If the library has stored the image of its company banner inside a constant, they will only have to make one change to the website code for it to be changed everywhere. Otherwise, they will have to dig through hundreds of files.

